MohammadReza Davari

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EDUCATION

PhD of Computer Science *Jan 2021 - May 2025*
Concordia University and Mila – Quebec AI Institute *Montreal, Canada*
Supervisor: Dr. Eugene Belilovsky
GPA: 4.00
Scholarships:
- Graduate Doctoral Incentive Fellowship *2021*
- Concordia University Conference and Exposition Award *2022*

Master of Computer Science *Sep 2017 - May 2020*
Concordia University *Montreal, Canada*
Supervisors: Dr. Leila Kosseim and Dr. Tien Bui
Thesis: Neural Network Approaches to Medical Toponym Recognition *Nominated for the best thesis award*
GPA: 3.95
Scholarships:
- Concordia Merit Scholarship *2018, 2019*
- Concordia University Conference and Exposition Award *2019*
- IVADO Student Grant *2020*

Bachelor of Science: Mathematics and Computer Science *Sep 2013 - Dec 2016*
McGill University *Montreal, Canada*
Represented McGill University in Putnam Mathematical Competition in both 2014 and 2015

College Diploma (DEC) in Pure and Applied Science *Sep 2011 - May 2013*
Dawson College *Montreal, Canada*
Dawson Mathematics Competition: 1st place in 2012 and 2nd place in 2011

WORK EXPERIENCE

Microsoft *Redmond, USA*

Senior Applied Scientist *Aug 2025 - Present*
- Built agentic evaluation pipelines for end-task product assessment, combining synthetic data generation, deterministic metrics, and LLM-as-judge evaluation to identify product-improvement opportunities
- Developed long-context and multimodal judge systems that ground evaluations in relevant document evidence, including agentic selection of only the pages/images actually used by product agents
- Shipped reusable internal judge SDK and reporting modules that convert evaluation results, telemetry, and logs into hierarchical failure analyses, root-cause insights, and actionable recommendations

Applied Scientist Intern *Jun 2024 - Aug 2024*
- Developed a novel method for LLM system prompt migration and optimization, achieving SOTA on multiple NLP benchmarks
- Implemented automatic LLM-based metric creation and metric evaluation given a target feature
- Observed product lift for prompt migration, reducing manual migration time

Concordia Applied AI Institute *Montreal, Canada*

Data Scientist *Nov 2023 - May 2024*
- Developed and helped deploy an NLP solution using BERT to automatically populate evaluation forms of long complex legal government contracts
- Implemented content highlighting to help users verify the context used by the model to reach a decision for each evaluation criterion
- Contributed to scoping new projects and writing NLP project proposals for new clients and feature requests

Eiffel MedTech *Montreal, Canada*

Machine Learning Project Contractor *Mar 2023 - Oct 2023*

- Developed an EOS X-ray image processing pipeline for 3D lower limb point cloud reconstruction, including bone and prosthetic segmentation
- Implemented fine-tuned Segment Anything Model (SAM) with manually labelled data for precise segmentation
- Created a Transformer-based neural network for 3D bone reconstruction from DRR and CT scan pairs. Deployed the pipeline on local servers for practical use

Amazon – Science *Vancouver, Canada*

Applied Research Scientist Intern *Jun 2022 - Sep 2022*

- Developed a novel method based on contrastive learning and self-prediction to learn universal user embeddings
- Applied the proposed user embedding to in-production downstream tasks and observed boost in performance
- The results were published in AMLC (Amazon internal Machine Learning Conference)

Ericsson *Montreal, Canada*

Machine Learning Technical Support Specialist *Sep 2021 - Feb 2022*

- Led the ML technical support of three projects: customer service chatbot, Python code completion, and sensitive log detection
- Carried out a series of NLP tutorials on the industrial applications of large scale language models
- Conducted workshops on data ingestion, processing, and visualisation

Ubisoft Divertissements Inc. *Montreal, Canada*

Data Scientist Intern *Sep 2021 - Feb 2022*

- Implemented hierarchical span-based toxicity detection model on in-game chat data
- Collaborated on implementing data annotation procedures

Coveo Solutions Inc. *Montreal, Canada*

Machine Learning Team Lead *Sep 2020 - Jun 2021*

- Led a group of 6 ML Scientist and Engineers on several NLP projects
- Collaborated with the product managers to plan road maps and communicate with customers/stakeholders
- Researched and implemented multi-task learning models to improve information flow between tasks and reduce resource consumption
- Researched and implemented an extractive Question Answering system for clients with minimally labelled data
- Adapted Continual Learning techniques to mitigate catastrophic forgetting and reduce training costs

Machine Learning Scientist *Sep 2019 - Sep 2020*

- Served ML models using TorchServe on Docker images in EMR
- Quantized and pruned the ML models to minimize the resources used and maximize throughput
- Implemented parallel and distributed model training on GPU
- Implemented a dynamic taxonomy prediction for queries, and used attention mechanism to gray-box the model
- Researched and implemented Transformer-based query retrieval system for long queries
- Implemented product content embedding based on product description and visual representation
- Participate in ML recruitment events and conduct technical interviews
- Initiated and organized *paper club*, a bi-weekly workshop on the recent developments in ML
- Represented Coveo at NLP events such as RALI, MILA, MTL-NLP, etc.

Machine Learning Developer Intern *May 2019 - Sep 2019*

- Implemented parallel model building on EMR clusters
- Implemented bigram and trigram model for product recommendation: frequently bought together recommender

TEACHING EXPERIENCE

Concordia University, Department of Computer Science *May 2018 - May 2019*
Teaching Assistant for: Artificial Intelligence, Combinatorics, *Montreal, Canada*
Theoretical Computer Science, Discrete Structures and Formal Languages

McGill University, Department of Mathematics *Sep 2015 - May 2016*
Teaching Assistant for: Analysis 1 and 2 *Montreal, Canada*

Dawson College *Sep 2011 - May 2016*
Teaching Assistant for: Linear Algebra, Probability and Statistics, Calculus 1, 2, and 3 *Montreal, Canada*

Liberty Tutoring *Sep 2013 - May 2016*
Private Tutor for: Linear Algebra, Probability and Statistics, Calculus 1, 2, and 3 *Montreal, Canada*
Waves and Optics, Mechanics, Electricity and Magnetism

SKILLS

Language Skills: Fluent in English and Persian; intermediate in French
Computer Skills: PyTorch, TensorServe, TensorFlow, Keras, Python, C/C++, Java, MATLAB, JavaScript, LaTeX, Bash, Scala, PySpark, Pandas, SQL, AWS, EMR, EC2, S3, Hadoop

RELEVANT COURSES

Self-supervised Learning (Université de Montreal)
Image Processing (Concordia University)
Natural Language Processing (McGill University)
Representation Learning (Université de Montreal)
Reinforcement Learning (McGill University)
Continual Learning (Université de Montreal)

PUBLICATIONS

1. **Model breadcrumbs: Scaling multi-task model merging with sparse masks**
MohammadReza Davari and Eugene Belilovsky, In European Conference on Computer Vision (**ECCV 2024**)

2. **Scalable Upcycling of Finetuned Foundation Models via Sparse Task Vectors Merging**
MohammadReza Davari and Eugene Belilovsky, In Foundation Models in the Wild Workshop (**ICML 2024 FM-Wild Workshop**)

3. **Prototype-Sample Relation Distillation: Towards Replay-Free Continual Learning**
Nader Asadi, MohammadReza Davari, Sudhir Mudur, Rahaf Aljundi and Eugene Belilovsky, In Proceedings of the International Conference on Machine Learning (**ICML 2023**)

4. **Reliability of CKA as a Similarity Measure in Deep Learning**
MohammadReza Davari, Stefan Horoi, Amine Natik, Guillaume Lajoie, Guy Wolf and Eugene Belilovsky, In Proceedings of the International Conference on Learning Representations (**ICLR 2023**)

5. **Deceiving the CKA Similarity Measure in Deep Learning**
MohammadReza Davari, Stefan Horoi, Amine Natik, Guillaume Lajoie, Guy Wolf and Eugene Belilovsky, In Machine Learning Safety Workshop (**NeurIPS ML Safety Workshop 2022**)

6. **Probing Representation Forgetting in Supervised and Unsupervised Continual Learning**
MohammadReza Davari, Nader Asadi, Sudhir Mudur, Rahaf Aljundi and Eugene Belilovsky, In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR 2022**)

7. **On the Inadequacy of CKA as a Measure of Similarity in Deep Learning**
MohammadReza Davari, Stefan Horoi, Amine Natik, Guillaume Lajoie, Guy Wolf and Eugene Belilovsky, In Geometrical and Topological Representation Learning Workshop (**ICLR GTRL Workshop 2022**)

8. **Probing Representation Forgetting in Continual Learning**
MohammadReza Davari and Eugene Belilovsky, In Distribution Shift Workshop (**NeurIPS DistShift Workshop 2021**)

9. **Semantic Similarity Matching Using Contextualized Representations**
Farhood Farahnak, Elham Mohammadi, MohammdReza Davari and Leila Kosseim, In Proceedings of the Canadian Conference on Artificial Intelligence (**CanAI 2021**)

10. **TIMBERT: Toponym Identifier For The Medical Domain Based on BERT**
MohammdReza Davari, Leila Kosseim and Tien Bui, In Proceedings of the International Conference on Computational Linguistics (**COLING 2020**)

11. **Toponym Identification in Epidemiology Articles - A Deep Learning Approach**
MohammdReza Davari, Leila Kosseim and Tien Bui, In Proceedings of the International Conference on Intelligent Text Processing and Computational Linguistics (**CICLing 2019**)
- Best poster award